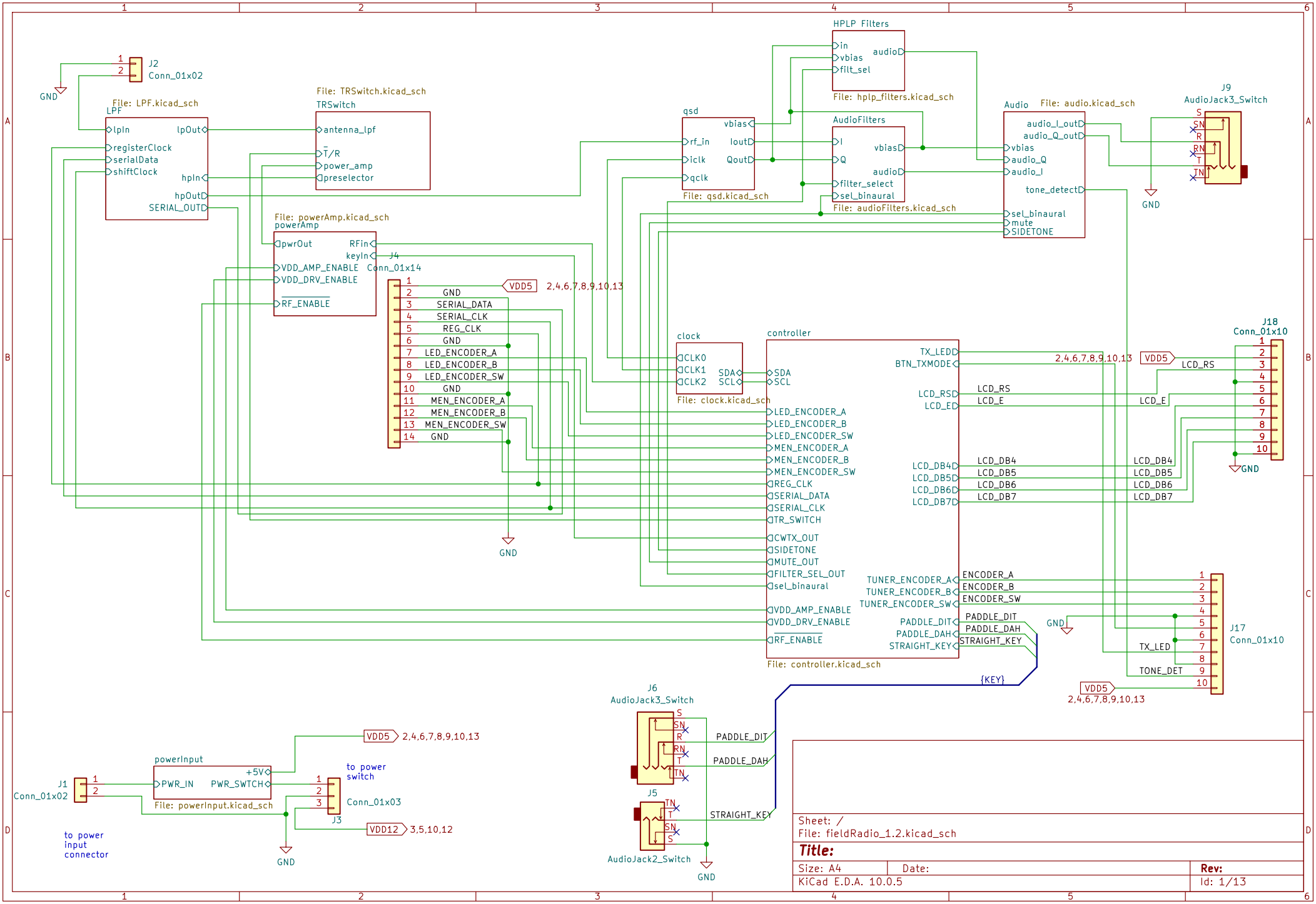
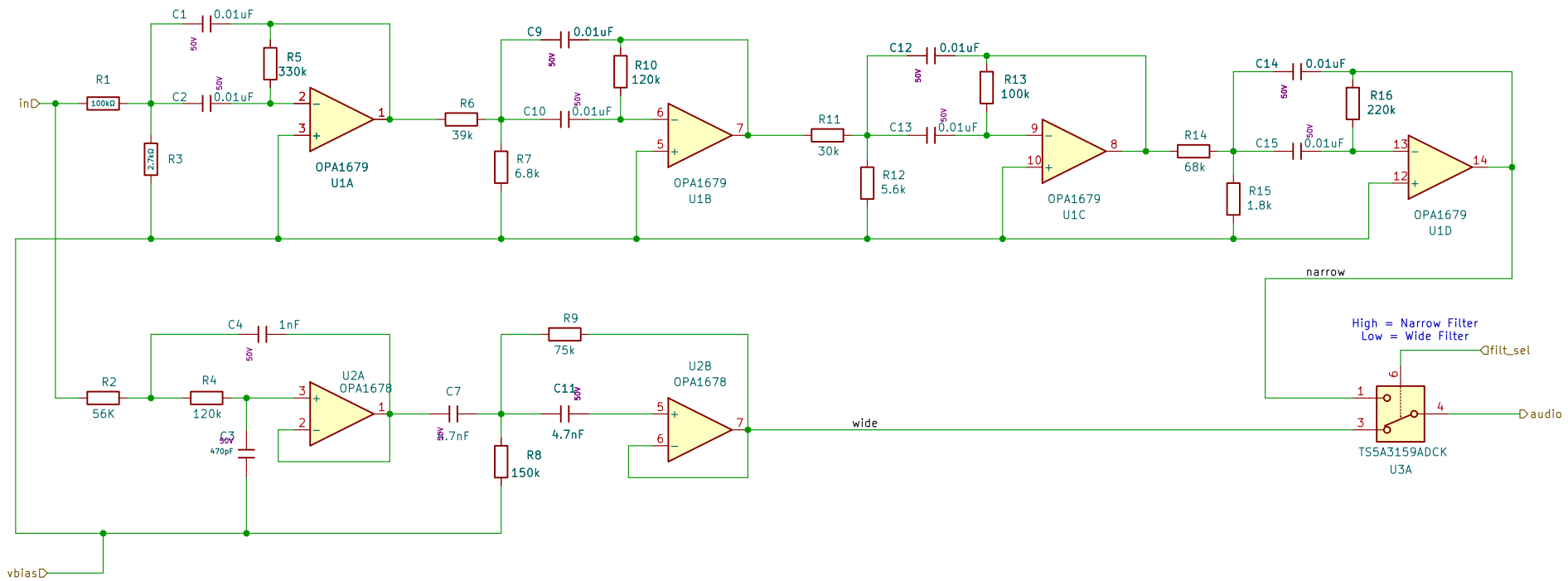
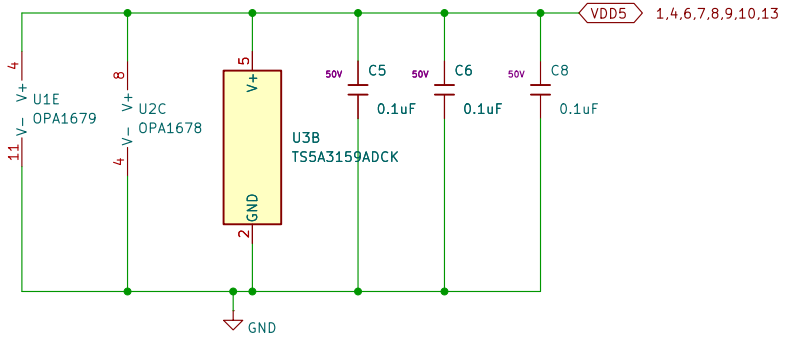

J2
Conn_01x02
GND
File: LPF.kicad_sch
LPF
lpIn
registerClock
serialData
shiftClock
lpOut
hpIn
hpOut
SERIAL_OUT
File: TRSwitch.kicad_sch
TRSwitch
antenna_lpf
T/R
power_amp
preselector
File: powerAmp.kicad_sch
powerAmp
pwrOut
RFin
keyIn
VDD_AMP_ENABLE
VDD_DRV_ENABLE
RF_ENABLE
J4
Conn_01x14
VDD5
2,4,6,7,8,9,10,13
GND
SERIAL_DATA
SERIAL_CLK
REG_CLK
LED_ENCODER_A
LED_ENCODER_B
LED_ENCODER_SW
MEN_ENCODER_A
MEN_ENCODER_B
MEN_ENCODER_SW
HPLP Filters
in
vbias
filt_sel
audio
File: hplp_filters.kicad_sch
qsd
rf_in
iclk
qclk
Iout
Qout
File: qsd.kicad_sch
AudioFilters
I
Q
filter_select
sel_binaural
File: audioFilters.kicad_sch
Audio
File: audio.kicad_sch
audio_I_out
audio_Q_out
audio_Q
audio_I
tone_detect
mute
SIDETONE
J9
AudioJack3_Switch
clock
CLK0
CLK1
CLK2
SDA
SCL
File: clock.kicad_sch
controller
TX_LEDD
BTN_TXMODE
LCD_RS
LCD_E
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
CWTX_OUT
MUTE_OUT
FILTER_SEL_OUT
TR_SWITCH
TUNER_ENCODER_A
TUNER_ENCODER_B
TUNER_ENCODER_SW
ENCODER_A
ENCODER_B
ENCODER_SW
PADDLE_DIT
PADDLE_DAH
STRAIGHT_KEY
{KEY}
File: controller.kicad_sch
J18
Conn_01x10
J17
Conn_01x10
TX_LED
TONE_DET
J6
AudioJack3_Switch
J5
AudioJack2_Switch
powerInput
PWR_IN
PWR_SWTCH
+5V
File: powerInput.kicad_sch
J1
Conn_01x02
J3
Conn_01x03
VDD12
3,5,10,12
to power switch
to power input connector
Sheet: /
File: fieldRadio_1.2.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/13



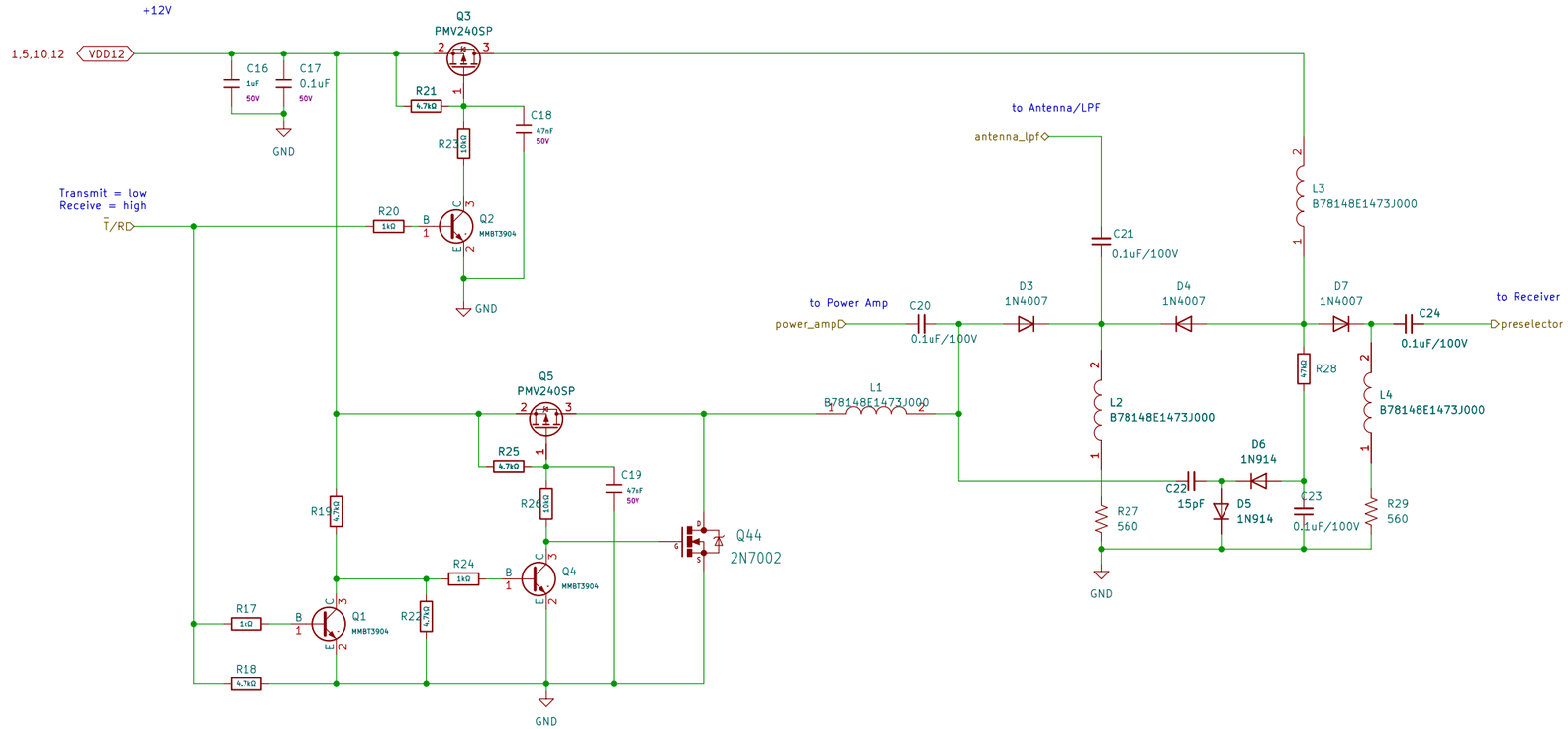


Wide/Narrow Audio Filters

Sheet: /HPLP Filters/
File: hplp_filters.kicad_sch

Title:

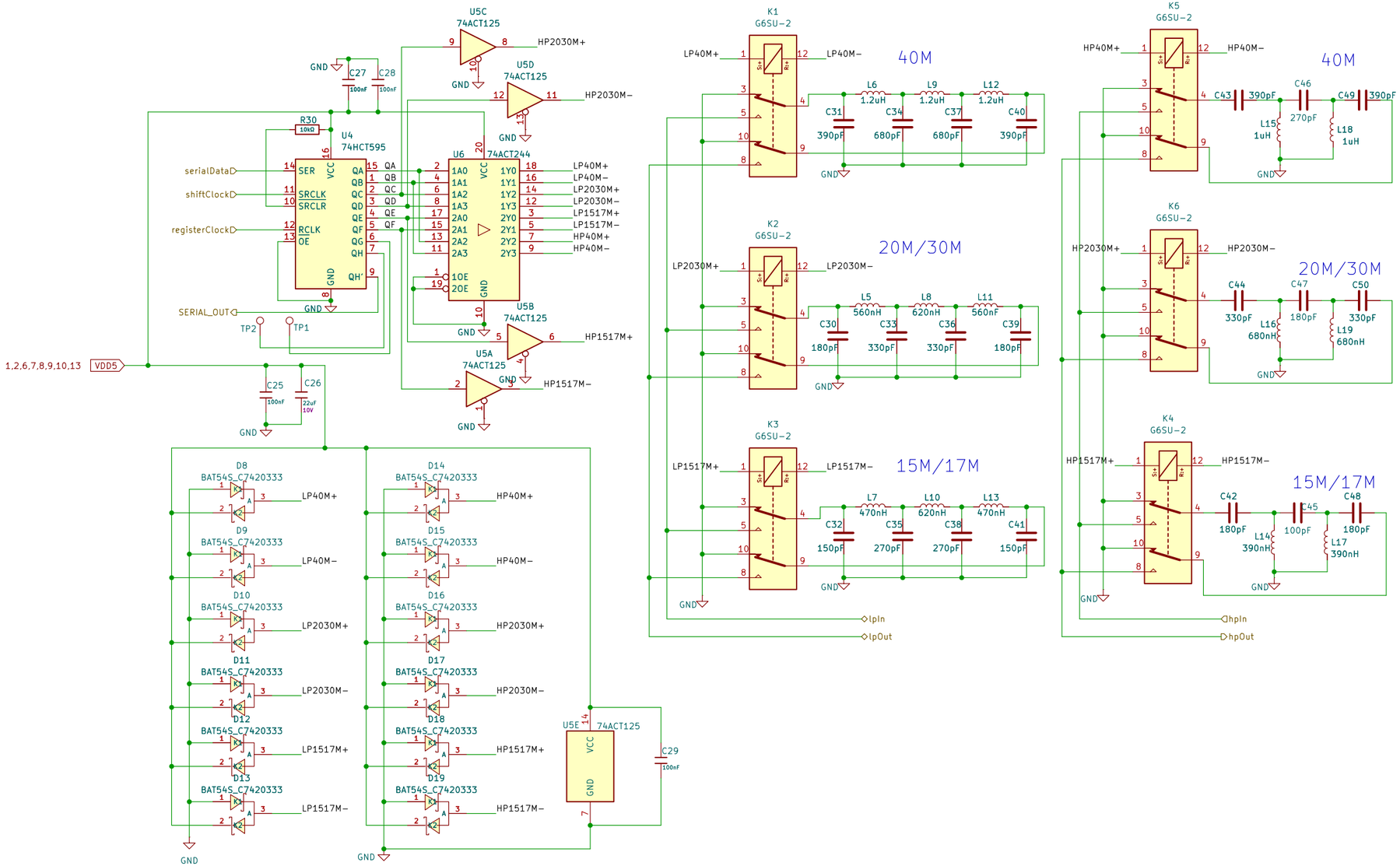
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/13



Sheet: /TRSwitch/
File: TRSwitch.kicad_sch

Title:

Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 3/13



Sheet: /LPF/		
File: LPF.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 4/13

Keyer Input
Active high

keyIn

R33 33kΩ

R35 8.2kΩ

R32 10kΩ

C55 220nF 25V

C56 100nF 50V

C57 100nF 50V

1,3,10,12 VDD12

U8A U8C OPA1678

R37 36kΩ

R38 10kΩ

R39 1kΩ

C61 220nF 25V

U8B OPA1678

C64 22nF 50V

k1 k2 k3 k4 k5 k6

VDD_AMP_ENABLED

U14 LMR51430

C75 4.7uF 25V

C77 4.7uF 25V

C78 100nF 50V

R47 100kΩ

C79 100nF 50V

sw1

L24 6.8uH

R50 10kΩ

fb1

R49 30kΩ

R48 0Ω

VDD_AMP

C83 22uF 25V

C85 22uF 25V

C86 22uF 25V

k9

R46 1kΩ

cm1 cm2

Q7A BC847BS

Q7B BC847BS

J8

to exernal
gain control

SW1 SW_Slide_DPDT

VAMP

L20 4.7uH

U7 TLV61047DDC

D20 B240

vbst

R36 150K

C58 10uF

C59 10uF

C60 1uF 50V

R31 10kΩ

R34 10kΩ

GND

VDD_DRV_ENABLED

U11 LMR51430

C65 4.7uF 25V

C66 4.7uF 25V

C67 100nF 50V

R41 10kΩ

C69 100nF 50V

sw2

L21 6.8uH

fb2

R45 100k

R43 12kΩ

R44 2kΩ

+5VA

C73 22uF 25V

C74 22uF 25V

C76 100uF

J13 Conn_01x02

VDD_PWR

C80 47uF

C82 10uF

1,3,10,12 VDD12

C51 47uF

C52 10uF

C53 10uF

C54 1uF 50V

GND

+3.3V

R40 10kΩ

U9B U9A 74LVC2G125

RF Input
3.3V square wave

RFin

rfd+ rfd−

RF_ENABLED

U12 UCC27517DBVR

U13 UCC27517DBVR

VDD GND IN+ IN− OUT

G1 G2

Q6B DMN10H220LPDW

Q6A DMN10H220LPDW

D1 D2

L22

L23

pwrOut

C81 100nF 100V

C68 100nF 50V

C70 100nF 50V

C71 2.2uF 50V

C72 2.2uF 50V

C62 100nF 50V

U9C 74LVC2G125

U10 MIC5235-3.3YM5

C63 100nF 50V

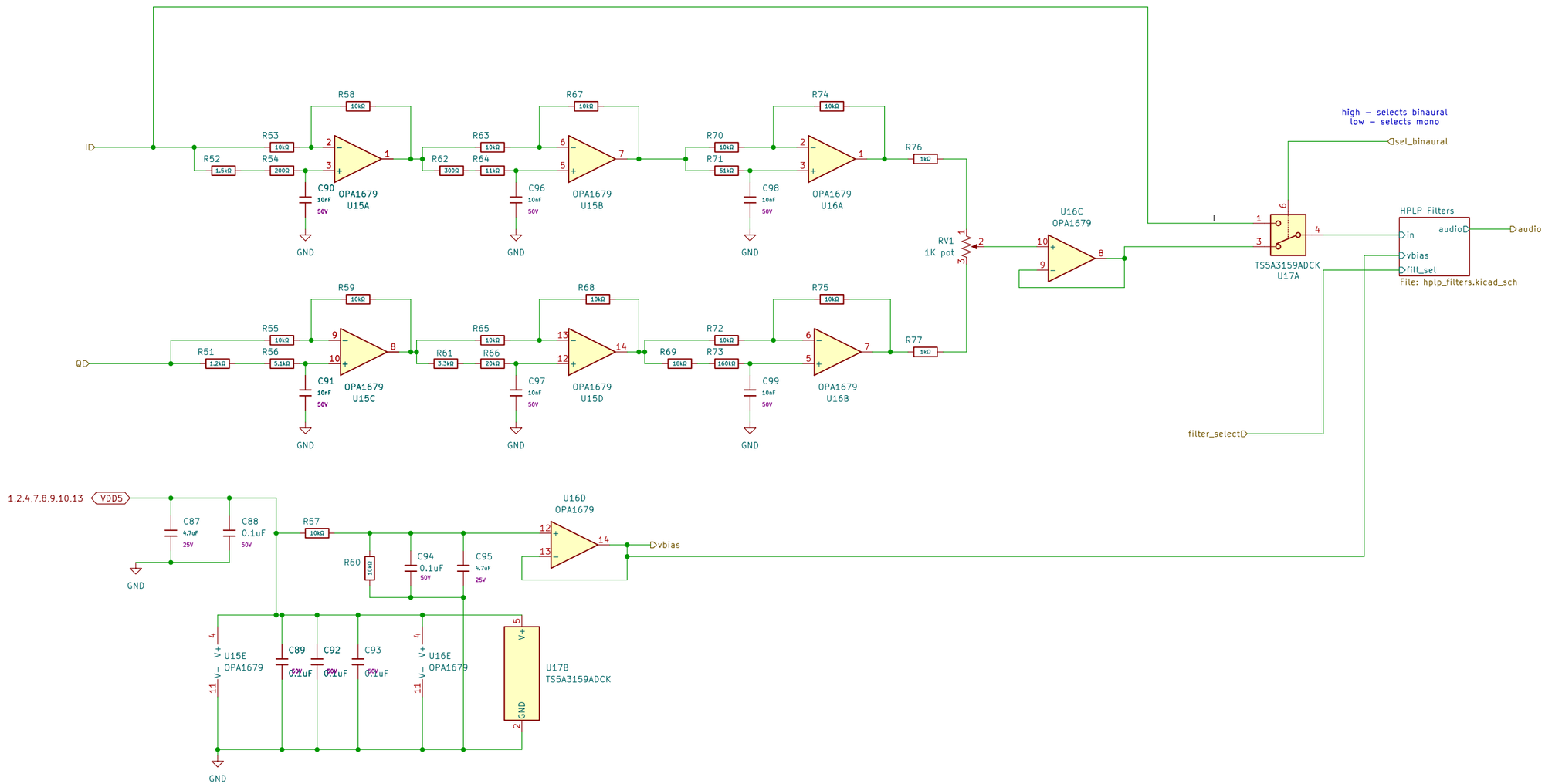
R42 1kΩ

Class D Amplifier Evaluation Board

Sheet: /powerAmp/
File: powerAmp.kicad_sch

Title:

Size: B | Date: | **Rev:**
KiCad E.D.A. 10.0.5 | Id: 5/13

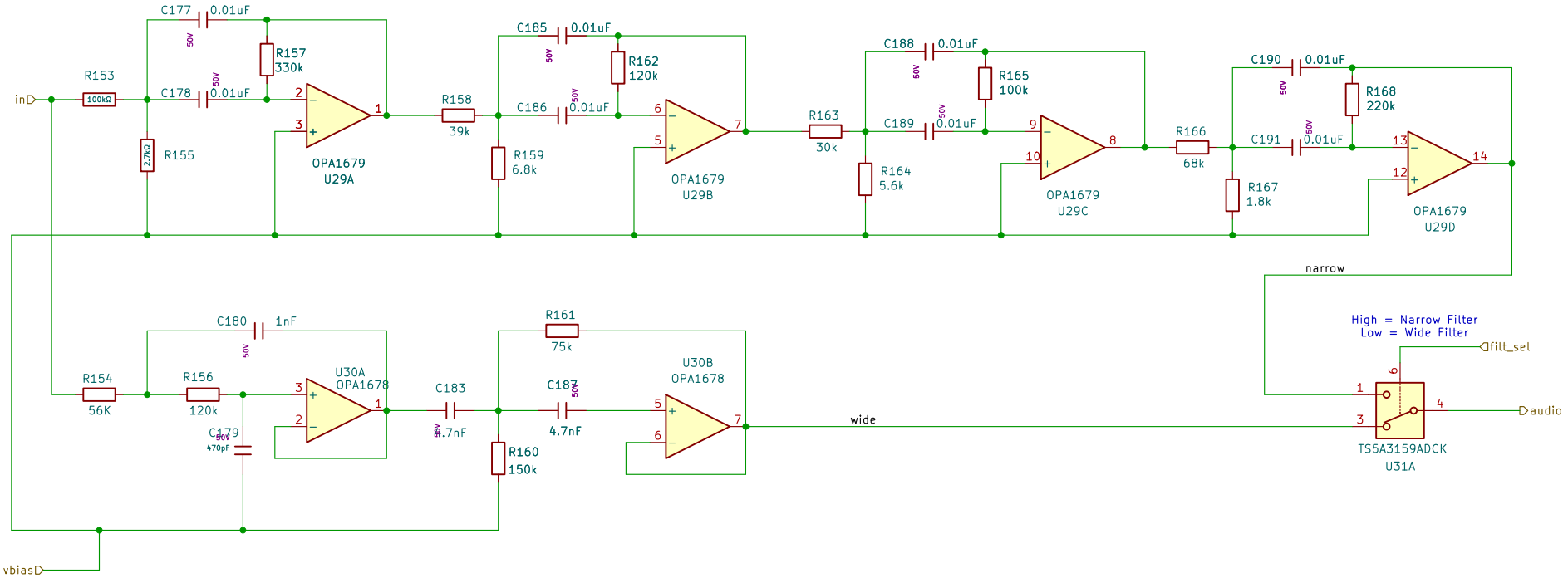


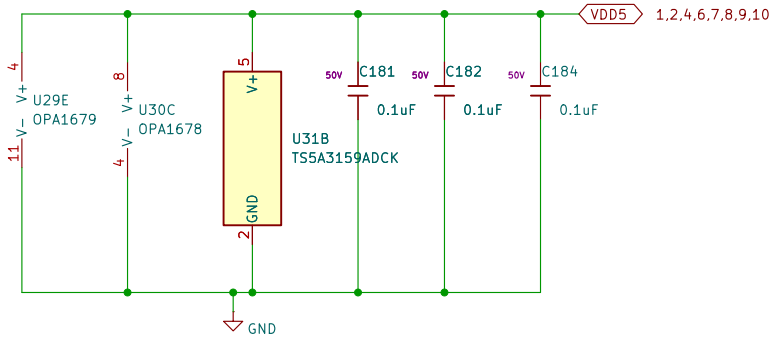
All Pass Filter and
Wide/Narrow Audio Filters

Sheet: /AudioFilters/
File: audioFilters.kicad_sch

Title:

Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 6/13





Wide/Narrow Audio Filters

Sheet: /AudioFilters/HPLP Filters/
File: hplp_filters.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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+5V supply

1,2,4,6,7,9,10,13 VDD5

C106 50V 0.1uF GND

R81 47kΩ

Sidetone

SIDETONE

C107 50V 0.1uF

R84 10kΩ

R85 10kΩ

R86 10kΩ

R82 47kΩ

C110 50V 0.1uF

C111 50V 0.1uF

C112 50V 0.1uF

U21A OPA1679

U21B OPA1679

C116 50V 0.1uF

C117 4.7uF 25V

GND

R93 10kΩ

R91 10kΩ

R90 10kΩ

R92 2.4kΩ

U21D OPA1679

C118 50V 0.1uF

C114 4.7uF 25V

R89 1kΩ

GND

J7 Conn_01x06

to 20K pot sidetone volume adjust

to 10K pot main volume adjust

R113 47kΩ

R114 47kΩ

MUTE = Active High

mute

U19A TS5A3159ADCK

audio_ID

R88 10kΩ

C115 4.7uF 25V

C113 4.7uF 25V

Audio inputs

U20A TS5A3159ADCK

audio_QD

R87 10kΩ

OPA1679 U21C

OPA1679 U21B

vbias

R83 10kΩ

C108 50V 0.1uF

C109 25V 4.7uF

GND

U22A TS5A3159ADCK

sel_binaural

high – selects binaural

low – selects mono

U22B TS5A3159ADCK

U20B TS5A3159ADCK

U19B TS5A3159ADCK

GND

toneDecoder

audio_in

detect

tone_detect

File: toneDecoder.kicad_sch

C121 50V 47uF

C124 0.1uF

GND

R96 20kΩ

C119 25V 470nF

R94 20kΩ

U23 LM4810

VDD VIN1 VIN2 SHDN VOUT1 VOUT2 BYPASS GND

C123 100uF

audio_I_out

C125 100uF

audio_Q_out

C120 25V 470nF

R95 20kΩ

C122 4.7uF 25V

R97 10kΩ

GND

R98 20kΩ

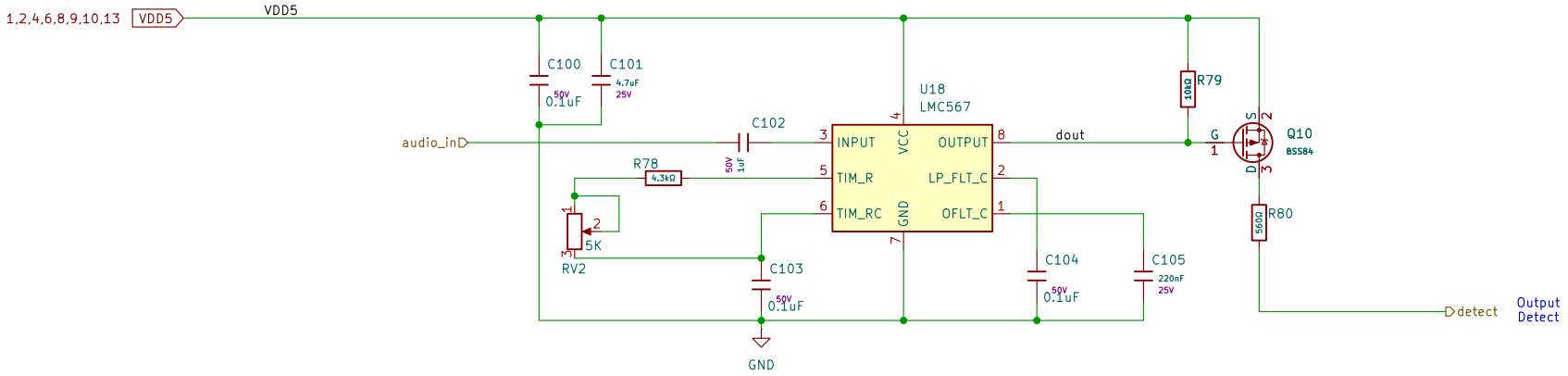
Sheet: /Audio/

File: audio.kicad_sch

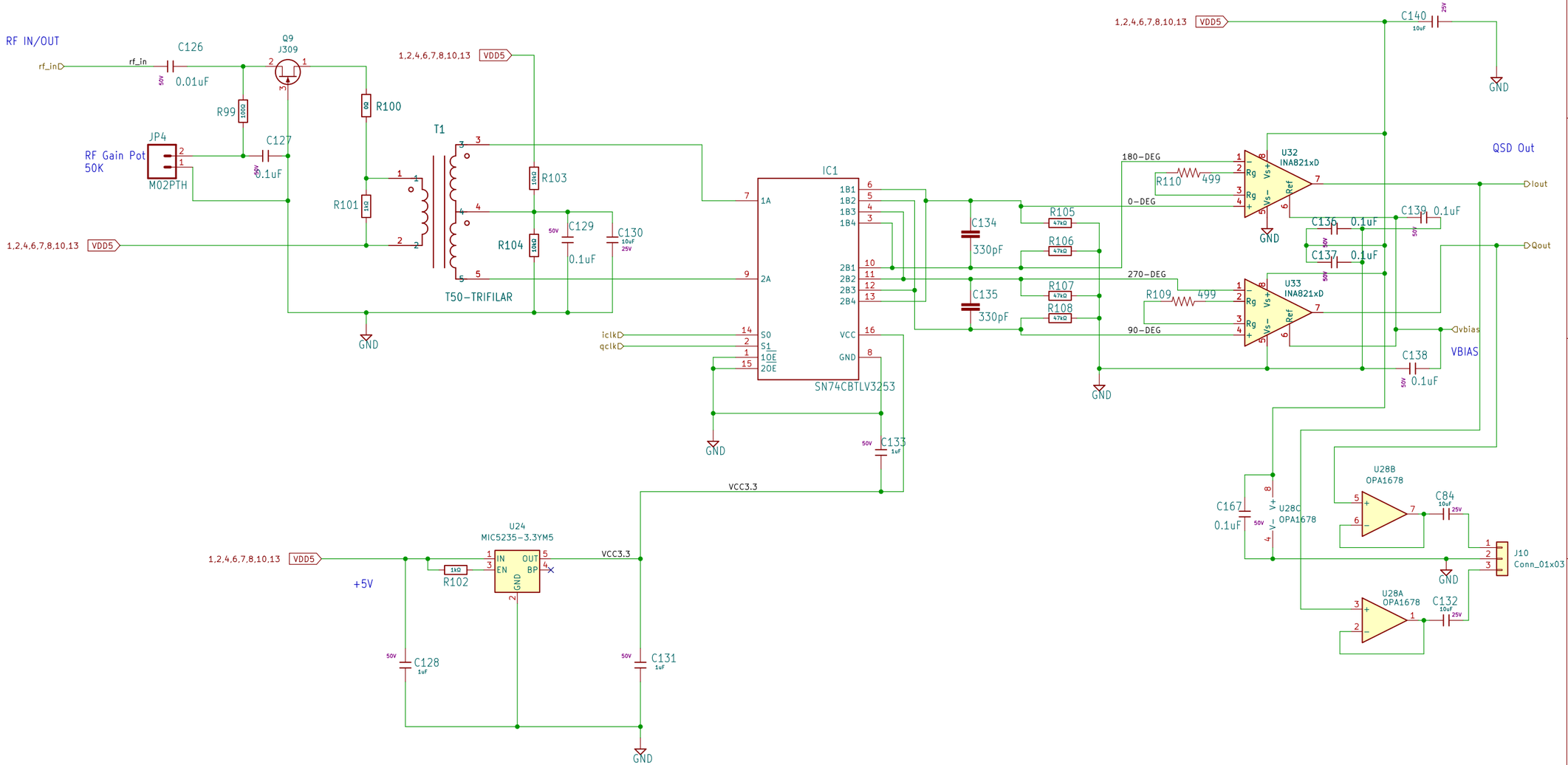
Title:

Size: B | Date: | **Rev:**

KiCad E.D.A. 10.0.5 | Id: 8/13

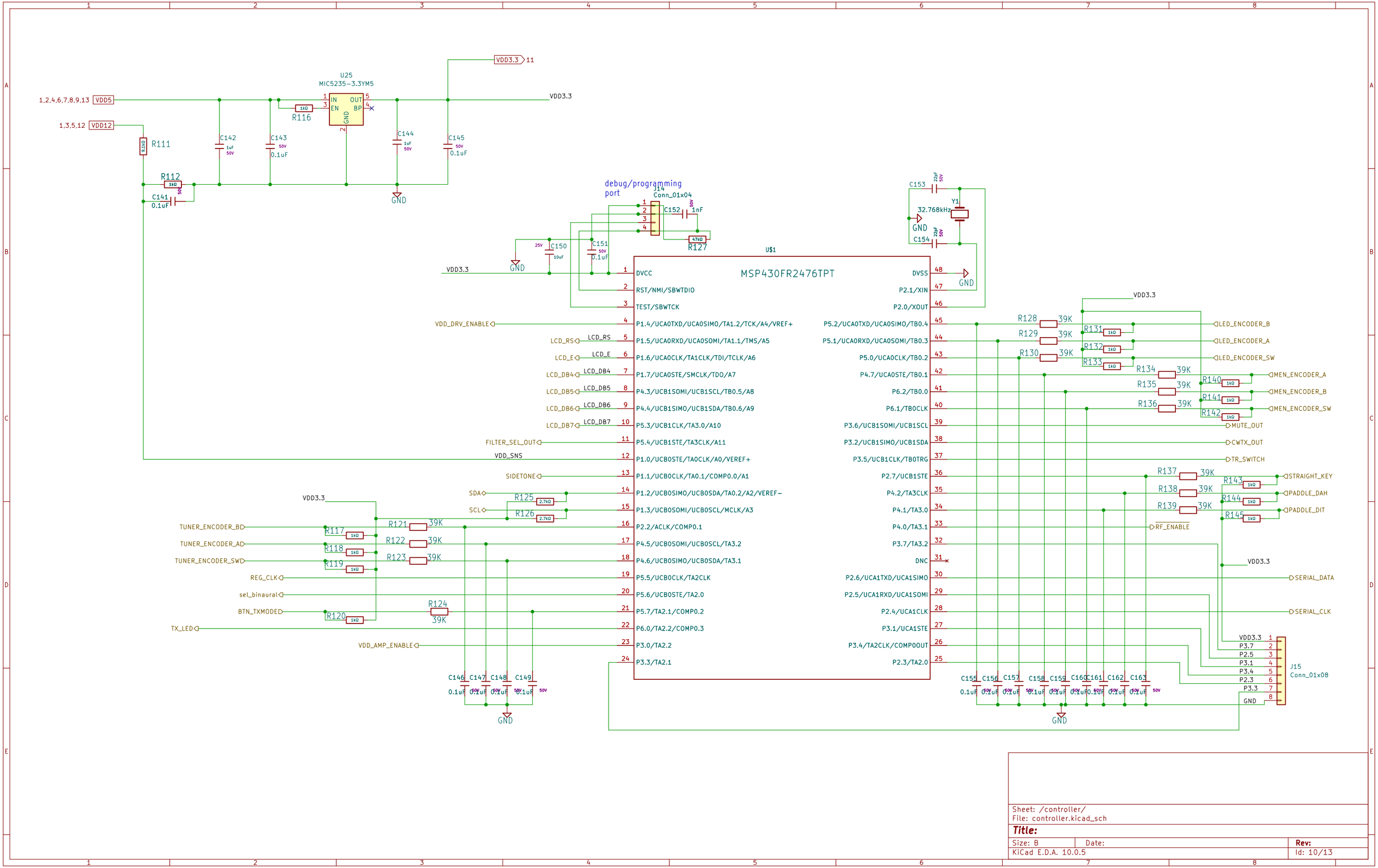


Sheet: /Audio/toneDecoder/		
File: toneDecoder.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 7/13



Quadrature Sampling Detector
Rev. C

Sheet: /qsd/		
File: qsd.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 9/13



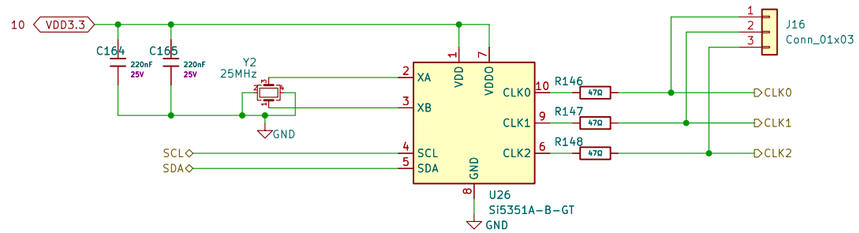
Sheet: /controller/
File: controller.kicad_sch

Title:

Size: B | Date:

KiCad E.D.A. 10.0.5

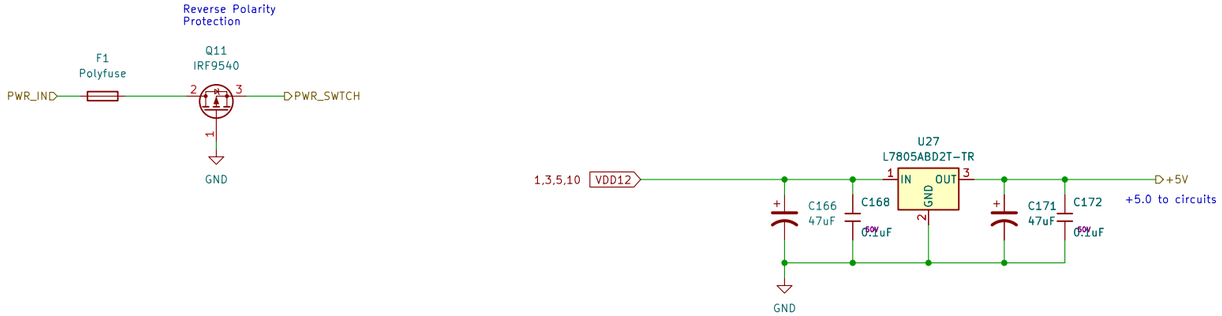
Rev: Id: 10/13



Sheet: /clock/
File: clock.kicad_sch

Title:

Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 11/13



Sheet: /powerInput/
File: powerInput.kicad_sch

Title:

Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 12/13